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Vincent Wang

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Let $N \trianglelefteq G$ and $H' \leq G/N$. Then there exists a unique $H \leq G$ such that $N \trianglelefteq H$ and $H/N = H'$.

Proof: Let $H = \{g \in G : \bar{g} \in H'\}$. Note that $1 \in H$ since $\bar{1} \in H'$. Suppose $h_1, h_2 \in H$. Then $\overline{h_1 h_2} = \bar{h_1} \cdot \bar{h_2} \in H'$ since H' is closed under $\cdot$, so $h_1 h_2 \in H$, ie H is closed under $\cdot$. Suppose $h \in H$. Then $\overline{h^{-1}} = \bar{h}^{-1} \in H'$, since H' is closed under -1 , so $h^{-1} \in H$, ie H is closed under -1 . Hence, $H \leq G$. Further, for all $n \in N$, $\bar{n} = \bar{1} \in H'$, so $N \leq H$.

By construction, if $h \in H$, $\bar{h} \in H'$, so $H/N \leq H'$. Conversely, suppose $\bar{h} \in H'$. Then $h \in H$, so $H/N = H'$.

Suppose H_1, H_2 contain N and $H_1/N = H_2/N$. Notice that $H_1 = \bigcup_{\bar{h} \in H_1/N} \bar{h}$ and similarly for H_2 , so $H_1 = H_2$.

Suppose further $H' \trianglelefteq G/N$. Then $H \trianglelefteq G$. Let $g \in G$ and $h \in H$. Then $\bar{h} \in H'$, so $\bar{g} \cdot \bar{h} \cdot \bar{g}^{-1} \in H'$ since $H' \trianglelefteq G/N$. It follows $\overline{ghg^{-1}} \in H'$, so $ghg^{-1} \in H$. Hence, $H \trianglelefteq G$.

Lemma. Let $\varphi : G \twoheadrightarrow H$ be an epimorphism and G be abelian. Then H is Abelian.

Proof: Let $h_1, h_2 \in H$. Since φ is surjective, let $h_1 = \varphi(g_1)$ and $h_2 = \varphi(g_2)$. Then $h_1 h_2 = \varphi(g_1) \varphi(g_2) = \varphi(g_1 g_2) = \varphi(g_2 g_1) = h_2 h_1$.

Lemma. Let $H \leq G$ and $K \trianglelefteq G$. Then $HK/K \cong H/(H \cap K)$.

Proof: Consider the surjective composition $\varphi : H \hookrightarrow HK \rightarrow HK/K$. Note $\ker \varphi = \{h \in H : hK = K\} = \{h \in H : h \in K\} = H \cap K$. Hence, $H \cap K \trianglelefteq H$ and the result follows by the First Isomorphism theorem.

Lemma. Let G be solvable and $N \trianglelefteq G$. Then G/N is solvable.

Proof. Let

$$1 = G_0 \trianglelefteq G_1 \trianglelefteq G_2 \trianglelefteq \dots \trianglelefteq G_s = G$$

where G_{i+1}/G_i is abelian for $i \in [[s-1]]$. Then

$$G_0N \trianglelefteq G_1N \trianglelefteq G_2N \trianglelefteq \dots \trianglelefteq G_sN,$$

so

$$G_0N/N \trianglelefteq G_1N/N \trianglelefteq G_2N/N \trianglelefteq \dots \trianglelefteq G_sN/N.$$

In particular, $G_0N/N = 1$ and $G_sN/N = G/N$. Further, by the Second Isomorphism theorem, $(G_{i+1}N/N)/(G_iN/N) \cong G_{i+1}N/G_iN$.

Consider the surjective composition $G_{i+1} \twoheadrightarrow G_{i+1}N \twoheadrightarrow G_{i+1}N/G_iN$, whose kernel contains G_i , so there exists a surjective map $G_{i+1}/G_i \rightarrow G_{i+1}N/G_iN$. Since G_{i+1}/G_i is abelian, so is $G_{i+1}N/G_iN$ by a previous lemma.

Lemma. Let $H, K \leq G$ and $N \trianglelefteq H$. Then $N \cap K \trianglelefteq H \cap K$. Further, $(H \cap K)/(N \cap K)$ is isomorphic to a subgroup of H/N .

Consider the composition $\varphi : H \cap K \hookrightarrow H \twoheadrightarrow H/N$. Suppose $x \in \ker \varphi$, so $hN = N$, ie $h \in N$. Hence, $x \in N \cap K$, since $N \subset H$. The result follows by the First Isomorphism theorem.

Lemma. Let G be solvable and $H \leq G$. Then H is also solvable.

Proof. Let

$$1 = G_0 \trianglelefteq G_1 \trianglelefteq G_2 \trianglelefteq \dots \trianglelefteq G_s = G,$$

where G_{i+1}/G_i is abelian for $i \in [[s-1]]$. Intersecting with H , we have

$$G_0 \cap H \trianglelefteq G_1 \cap H \trianglelefteq G_2 \cap H \trianglelefteq \dots \trianglelefteq G_s \cap H.$$

Note $G_0 \cap H = 1$ and $G_s \cap H = H$. Further, $(G_{i+1} \cap H)/(G_i \cap H)$ is isomorphic to a subgroup of G_{i+1}/G_i , which is abelian. Subgroups of abelian groups are abelian, so the result follows.

Theorem. Together, let $N \trianglelefteq G$. Then G is solvable iff N and G/N are solvable (note we weaken one result since we only use that $N \leq G$ in one direction).

Lemma. Let G be solvable. Then there exists a chain

$$1 = G_0 \triangleleft G_1 \triangleleft G_2 \triangleleft \dots \triangleleft G_s = G,$$

where $G_{i+1}/G_i \cong \mathbb{Z}_p$ for a prime p for all $i \in [[s-1]]$.

Proof: Let G be solvable. Suppose $N \triangleleft G$ with G/N abelian and $|G/N| \neq p$

for a prime p . Let $p \mid |G/N|$. Since G/N is abelian, pick $H' \leq G/N$ such that $|H'| = |G/N|/p$. Let H' correspond to the group H with $H' = H/N$. Then

$$|G/H| = |(G/N)/(H/N)| = |(G/N)/H'| = p.$$

Since G/N is abelian, $H' \trianglelefteq G/N$, so we in fact have $H \trianglelefteq G$. $N \triangleleft H$ since $|G/N| \neq p$ (I stress that equality is not achievable). Hence, we have the chain $N \triangleleft H \triangleleft G$. By induction on $|G|$, the result follows.

Lemma. Let $G \neq 1$ be solvable and simple. Then $G \cong \mathbb{Z}_p$ for a prime p .

Proof: We have G is abelian. Let $p \mid |G|$ and $x \in G$ have $|x| = p$. If $|G| > p$, then $1 \triangleleft \langle x \rangle \triangleleft G$, absurd. Hence, $|G| = p$, for which $G \cong \mathbb{Z}_p$.

Lemma. Let G be solvable. Then the composition factors of G must have prime order.

Proof: Let $N \triangleleft G$ such that G/N is simple. Since G is solvable, so is G/N . Simple and solvable implies isomorphism to $\mathbb{Z}_p$. Subgroups of solvable groups are also solvable. By induction on $|G|$, the result follows.

Lemma. Let $H \leq G$ and $N = \bigcap_{g \in G} gHg^{-1}$. Then $N \trianglelefteq G$.

Proof: Let $x \in N$, so $g_1^{-1}xg_1 \in H$ for all $g_1 \in G$. Let $g_2 \in G$ and consider $y = g_2^{-1}xg_2$. Let $g_3 \in G$ be arbitrary, so

$$g_3^{-1}yg_3 = g_3^{-1}g_2^{-1}xg_2g_3 = (g_2g_3)^{-1}x(g_2g_3) \in H,$$

so $y = g_2^{-1}xg_2 \in N$, ie $N \trianglelefteq G$.

Lemma. Let G be solvable and $1 \triangleleft M \trianglelefteq G$ be minimal. Then M is abelian.

Proof: Pick $N \triangleleft M$ such that $[M : N] = p$ for a prime p . Then $M/N \cong \mathbb{Z}_p$. Consider the map $M \rightarrow M/N$. Since M/N is abelian, we descend to $M/[M, M] \rightarrow M/N$, so $[M, M] \leq N$.

Now let $g \in G$ be arbitrary. We have $gNg^{-1} \triangleleft gMg^{-1} = M$ of index p , so we have $[M, M] \leq gNg^{-1}$. Hence, $[M, M] \leq \bigcap_{g \in G} gNg^{-1} = M'$.

By the previous lemma, $M' \trianglelefteq G$. Note that $M' \leq N < M$, so $M' < M$. By minimality of M , $M' = 1$, for which $[M, M] = 1$, ie M is abelian.

Theorem. Let G be solvable. Then there exists a chain

$$1 = G_0 \triangleleft G_1 \triangleleft G_2 \triangleleft \dots \triangleleft G_s = G,$$

where G_{i+1}/G_i is abelian and $G_i \triangleleft G$.

Proof: We induct on $|G|$. If $|G|$ is simple, then $G \cong \mathbb{Z}_p$, and we're done. Else, there exists a minimal normal nontrivial subgroup M of G . By the previous lemma, M is abelian. We have G/M is simple. Let

$$\bar{1} = G'_1 \triangleleft G'_2 \triangleleft G'_3 \triangleleft \dots \triangleleft G'_s = G/M$$

be given by the IH. By the lattice theorem, let $G'_i = G_i/M$, so

$$M = G_1 \triangleleft G_2 \triangleleft G_3 \triangleleft \dots \triangleleft G_s = G.$$

Combining with $G_0 = 1$, we complete the chain. This chain is valid since $G'_i \triangleleft G/M \rightarrow G_i \triangleleft G$ by the Lattice theorem and $G_{i+1}/G_i \cong G'_{i+1}/G'_i$ by the Second Isomorphism theorem.

Let p be a prime, $a > 1$, and P be a group of order p^a . Then P is nilpotent with class at most $a - 1$.

I claim for $0 \leq n \leq a$ that $|Z_n(P)| \geq p^n$. I induct on n . The result holds for $n = 0$. Suppose the result for n . If $Z_n(P) = P$, then $Z_{n+1}(P) = P$, so $|Z_{n+1}(P)| = p^a \geq p^{n+1}$. Else, $Z_n(P) < P$, so $P/Z_n(P) \neq 1$, so $Z_{n+1}(P)/Z_n(P) = Z(P/Z_n(P)) \neq 1$, so $|Z_{n+1}(P)| = |Z_{n+1}(P)/Z_n(P)| \cdot |Z_n(P)| \geq p \cdot p^n = p^{n+1}$.

In particular, $|Z_{a-2}(P)| \geq p^{a-2}$, so $|P/Z_{a-2}(P)| \leq p^2$, so $P/Z_{a-2}(P)$ is abelian. It follows $Z_{a-1}(P) = P$, so P is nilpotent of height at most $a - 1$.

Let G be finite with $p_1, p_2, \dots, p_s$ as the distinct primes dividing $|G|$. Let $P_i \in \text{Syl}_{p_i}(G)$.

If $P_i \trianglelefteq G$, then $G \cong \prod_{i=1}^s P_i$ by induction.

Let G_1, G_2 be groups. I claim that $Z(G_1 \times G_2) = Z(G_1) \times Z(G_2)$.

Proof: Suppose $(z_1, z_2) \in Z(G_1 \times G_2)$. Then $(z_1, z_2)(g_1, 1) = (g_1, 1)(z_1, z_2)$ for all $g_1 \in G_1$, ie $z_1 g_1 = g_1 z_1$, so $z_1 \in Z(G_1)$. Similarly, $z_2 \in Z(G_2)$, so $(z_1, z_2) \in Z(G_1) \times Z(G_2)$. Conversely, if $(z_1, z_2) \in Z(G_1) \times Z(G_2)$, then for all $(g_1, g_2) \in G_1 \times G_2$, we have

$$(z_1, z_2)(g_1, g_2) = (z_1 g_1, z_2 g_2) = (g_1 z_1, g_2 z_2) = (g_1, g_2)(z_1, z_2),$$

so $(z_1, z_2) \in Z(G_1 \times G_2)$.

Note that if $N_1 \trianglelefteq G_1, N_2 \trianglelefteq G_2$, then $(G_1 \times G_2)/(N_1 \times N_2) \cong (G_1/N_1) \times (G_2/N_2)$.

I claim that $Z_j(G) = \prod_{i=1}^s Z_j(P_i)$ for all $j \geq 0$. I induct on j . For $j = 0$, the result holds. Sps the result for j and let $x_i \in Z_{j+1}(P_i)$, $g_i \in P_i$. Let $x = (x_i)_{i \in [s]}$ and $g = (g_i)_{i \in [s]}$. Then

$$[x, g] = ([x_i, g_i])_{i \in [s]} \in \prod_{i=1}^s Z_j(P_i) = Z_j(G),$$

so $x \in Z_{j+1}(G)$.

Conversely, sps $x \in Z_{j+1}(G)$. Let $g \in G$ be arbitrary. Then

$$[x, g] = ([x_i, g_i])_{i \in [s]} \in Z_j(G) = \prod_{i=1}^s Z_j(P_i),$$

so $[x_i, g_i] \in Z_j(P_i)$ for all $i \in [s]$, so $x_i \in Z_{j+1}(P_i)$, so $x \in \prod_{i=1}^s Z_{j+1}(P_i)$, completing the induction.

Since P_i is a p group, P_i is nilpotent, say of height c_i . Let $c = \max_{i \in [s]} c_i$. Then $Z_c(G) = \prod_{i=1}^s Z_c(P_i) = \prod_{i=1}^s P_i = G$, so G is nilpotent of height at most c .

Suppose $N \leq H < G$ and $N \triangleleft G$. If $\bar{x} \in \bar{H}$, then $x \in H$.

Proof: $\bar{x} = \bar{h}$ for some $h \in H$, so $\overline{xh^{-1}} = \bar{1}$, so $xh^{-1} \in N$, so $x \in NH = H$, as desired.

Let P be a group with order p^a , $a \geq 1$. Suppose $H < P$. Then $H < N_P(H)$.

Proof: I induct on a . If $a = 1, 2$, then P is abelian, so $N_P(H) = P$ for all $H < P$.

Note $Z(P) \neq 1$. If $Z(P) \not\leq H$, then $H < \langle H, Z(P) \rangle \leq N_P(H)$, AWD. Else, $Z(P) \leq H$. Let $\bar{H} = H/Z(P)$ and $\bar{P} = P/Z(P)$. Note that $|\bar{P}| < |P|$ since $Z(P) \neq 1$. Since $\bar{H} < \bar{P}$, by IH, $\bar{H} < N_{\bar{P}}(\bar{H})$. I claim that $N_{\bar{P}}(\bar{H}) = N_P(H)/Z(P)$.

Suppose $\bar{x} \in N_{\bar{P}}(\bar{H})$, so for all $\bar{h} \in \bar{H}$, we have $\bar{x}\bar{h}\bar{x}^{-1} = \overline{xhx^{-1}} \in \bar{H}$, so $xhx^{-1} \in H$, so $x \in N_P(H)$.

Conversely, suppose $x \in N_P(H)$. Then for all $h \in H$, we have $xhx^{-1} \in H$, so $\bar{x}\bar{h}\bar{x}^{-1} \in \bar{H}$, so $\bar{x} \in \bar{H}$.

We have $H/Z(P) < N_P(H)/Z(P)$, so $H < N_P(H)$.

Let G be nilpotent and $N \leq G$. Then I claim G/N is nilpotent.

Specifically, let $\pi : G \rightarrow G/N$ be the projection map. Then $Z_n(G/N) \geq \pi(Z_n(G))$.

I induct on n . For $n = 0$, $Z_0(G/N) = \bar{1}$ and $\pi(Z_0(G)) = \pi(1) = \bar{1}$. Suppose the result for n . Let $x \in Z_{n+1}(G)$, so $[x, y] \in Z_n(G)$ for all $y \in G$, so $[\pi(x), \pi(y)] \in \pi(Z_n(G)) \leq Z_n(G/N)$. Since π is surjective, $[\pi(x), z] \in Z_n(G/N)$ for all $z \in G/N$, so $\pi(x) \in Z_{n+1}(G/N)$, completing the induction.

It follows that if G is of nilpotent class c , then G/N is of nilpotent class at most c .

Let G be finite nilpotent. Then for all $H < G$, we have $H < N_G(H)$.

Proof: We induct on the nilpotent class c of G . If $c = 1$, then G is abelian, so $N_G(H) = G > H$. Else, if $Z(G) \not\leq H$, then $H < \langle H, Z(G) \rangle \leq N_G(H)$, so assume $Z(G) \leq H$. Consider $\bar{G} = G/Z(G)$ and $\bar{H} = H/Z(G)$.

By a similar induction argument, $Z_n(G/Z(G)) \geq \pi(Z_{n+1}(G))$ for all $n \geq 0$, where $\pi : G \rightarrow G/Z(G)$ is the projection. Hence, $G/Z(G)$ is of nilpotent class at most $c-1$ (in fact proper containment implies that the class is equal to $c-1$). Since $\bar{H} < \bar{G}$, by the IH, we have $\bar{H} < N_{\bar{G}}(\bar{H})$. By the lattice theorem, we have $H < N_G(H)$, as desired.

Suppose $P \in \text{Syl}_p(G)$ for some prime $p \mid |G|$. Then $P \trianglelefteq G$. Let $N = N_G(P)$, so $P \trianglelefteq N$. Since $v_p(|P|) = v_p(|N|)$, $P \text{ char } N$. Since $N \trianglelefteq N_G(N)$, we have $P \trianglelefteq N_G(N)$. By maximality of N , we have $N_G(N) \leq N$, so $N_G(N) = N$. If $N < G$, then $N < N_G(N)$, absurd. It follows $N = G$, ie $P \trianglelefteq G$.

Let G be finite st for all $n \in \mathbb{Z}^+$ with $n \mid |G|$, there are at most n solutions to $x^n = 1$. Then G is cyclic.

Let $|G| = p^a r$ with $p \nmid r$. Then $S = \{x \in G : x^{p^a} = 1\} = \bigcup_{P \in \text{Syl}_p(G)} P$. Pick $P \in \text{Syl}_p(G)$, so $S \supset P$, so $p^a \geq |S| \geq p^a \rightarrow |S| = p^a \rightarrow S = P$. In particular, $P \trianglelefteq G$, so P is nilpotent, and of the form $P \cong \prod_{i=1}^s P_i$ for p -groups P_i of distinct primes.

Let $P = P_i$ for some i . By Lagrange, for $0 \leq k \leq a$, the number of solutions to $x^{p^k} = 1$ in G equals the number of solutions to the same equation in P . Suppose FTSOC $P^{p^k} = 1$ for some $0 \leq k < a$. Then the $p^a > p^k$ elements of P solve $x^{p^k} = 1$, absurd. Hence, there exists $x \in P$ with order p^a , ie P is cyclic. By Chinese remainder theorem, G is cyclic. The converse is true.

Define $p : H \rtimes K \rightarrow HK$ by $p(h, k) = hk$. I claim that p is a homomorphism. Indeed

$$\begin{aligned} p((h_1, k_1)(h_2, k_2)) &= p(h_1 k_1 \cdot h_2, k_1 k_2) = h_1 k_1 \cdot h_2 k_1 k_2 \\ &= h_1 k_1 h_2 k_1^{-1} k_1 k_2 = h_1 k_2 h_2 k_2 = p(h_1, k_1) p(h_2, k_2). \end{aligned}$$

If $p(h, k) = hk = 1$, then $h = k^{-1} \in H \cap K$, so $(h, k) = (1, 1)$, so $\ker p = 1$. p is clearly surjective, so p is an isomorphism. If $G = HK$, then $G \cong H \rtimes K$.

Let $p < q$ be primes and G be a group with order pq . Then $G \cong Q \rtimes P$, where P, Q are cyclic p, q groups, respectively.

Proof: $n_q \equiv 1 \pmod{q}$ and $n_q \mid p$, so $n_q = 1$. Hence, for $Q \in \text{Syl}_q(G)$, $Q \triangleleft G$. Let $P \in \text{Syl}_p(G)$. Note that $P \cap Q = 1$ since p, q are coprime. Note $QP \cong Q \rtimes P$ has order pq , so $G \cong Q \rtimes P$.

Consider $\text{Aut}(Q) \cong \mathbb{Z}_{q-1}$. If $p \nmid q-1$, then $\varphi : P \rightarrow \text{Aut}(Q)$ is trivial, for which $G \cong Q \times P \cong \mathbb{Z}_{pq}$. Else, $p \mid q-1$. Assume φ is nontrivial. Let $P = \langle x \rangle$. Let $\langle \gamma \rangle \leq \text{Aut}(Q)$ denote the unique subgroup of order p (since $\text{Aut}(Q)$ is cyclic). Let $\varphi_i(x) = \gamma^i$ for $0 \leq i < p$. If $i = 0$, then φ_i is trivial, for which $G \cong \mathbb{Z}_{pq}$. Else, there exists a generator x' such that $\varphi_i(x') = \gamma$. Since the choice of the generator of P is arbitrary, these groups are isomorphic.

Let G be a group of order 30. Then $n_5 \equiv 1 \pmod{5}$ and $n_5 \mid 6$